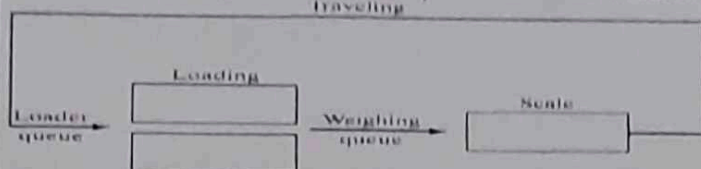




Semester: July 2025 –November 2025		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01		Semester: VII (SVU 2020)
Programme: Computer Engineering		
Class: LY B. Tech		
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: COMP
Course Code: 116U01E744	Name of the Course: Computer Simulation and Modeling	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks																				
Q1	Solve any FOUR	20																				
i)	Identify when simulation is appropriate and/or simulation is not needed. Justify the answer.	5																				
ii)	For a banking system simulation, identify at least 2 Entities, 2 Activities, 2 Events and 2 State Variables. Justify the answer.	5																				
iii)	Differentiate w.r.t simulation and modelling : Discrete Vs Continuous systems	5																				
iv)	Differentiate w.r.t simulation and modelling : Deterministic Vs Stochastic model	5																				
v)	What are 3 types of world views of the model used in simulation?	5																				
vi)	Draw flow chart of Departure Event in Event scheduling algorithm. Explain the same.	5																				
Q2 A	Solve the following	10																				
i)	Consider a Refrigerator selling company. The inventory is maintained as by principle of order-up-to-level (M) with review period of N days. The demand is distributed as shown in table. <table border="1"> <thead> <tr> <th>Demand</th> <th>Probability</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0.1</td> </tr> <tr> <td>1</td> <td>0.25</td> </tr> <tr> <td>2</td> <td>0.4</td> </tr> <tr> <td>3</td> <td>0.2</td> </tr> <tr> <td>4</td> <td>0.05</td> </tr> </tbody> </table> The distribution of lead time is as following <table border="1"> <thead> <tr> <th>Lead Time</th> <th>Probability</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.4</td> </tr> <tr> <td>2</td> <td>0.3</td> </tr> <tr> <td>3</td> <td>0.3</td> </tr> </tbody> </table> M = 10 and N = 4 days Let simulation start with inventory of 5 refrigerators and order of 5 refrigerators is expected to arrive in 3 days. Random digits for demand: 34, 59, 12, 09, 37, 29, 73, 61, 45 Random digits for lead time: 6, 3, 8, 1, 2, 9, 5, 6 Generate simulation table for 10 days period consisting of 2 review cycles.	Demand	Probability	0	0.1	1	0.25	2	0.4	3	0.2	4	0.05	Lead Time	Probability	1	0.4	2	0.3	3	0.3	05
Demand	Probability																					
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1	0.4																					
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ii)	Find out how well the current (M, N) system is working with respect to shortage of refrigerators in the 02 review cycles. Justify the answer.	05																				

OR																				
Q2 A	Consider following schema of Dump Truck Problem. Assume 4 Dump trucks.  Simulate using Event Scheduling algorithm till time 30. Consider following random times for activities. <table><tr><th>Loading Times</th><th>Weighing Time</th><th>Travel Time</th></tr><tr><td>10</td><td>12</td><td>60</td></tr><tr><td>5</td><td>12</td><td>40</td></tr><tr><td>5</td><td>16</td><td>40</td></tr><tr><td>10</td><td>12</td><td></td></tr></table>	Loading Times	Weighing Time	Travel Time	10	12	60	5	12	40	5	16	40	10	12		10			
Loading Times	Weighing Time	Travel Time																		
10	12	60																		
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10	12																			
Q 2 B	Solve any ONE	10																		
i)	Explain Event scheduling algorithm. Write all steps. What is importance of Future Event List in this algorithm?	10																		
ii)	People arrive at the Telephone booth with inter-arrival times distributed as follows, <table><tr><th>Inter-arrival time</th><th>Probability</th></tr><tr><td>5</td><td>0.2</td></tr><tr><td>6</td><td>0.7</td></tr><tr><td>7</td><td>0.1</td></tr></table> The call duration distribution is as follows <table><tr><th>Call duration</th><th>Probability</th></tr><tr><td>4</td><td>0.15</td></tr><tr><td>5</td><td>0.45</td></tr><tr><td>6</td><td>0.30</td></tr><tr><td>7</td><td>0.10</td></tr></table> Random digits for inter-arrival times: 6, 8, 3, 9, 2, 3, 7, 1, 4, 5 Random digits for call duration: 23, 45, 67, 12, 89, 47, 53, 71, 39, 94 Simulate the system for 10 arrivals. Comment on the need for addition a telephone booth on the basis of waiting time of the callers.	Inter-arrival time	Probability	5	0.2	6	0.7	7	0.1	Call duration	Probability	4	0.15	5	0.45	6	0.30	7	0.10	10
Inter-arrival time	Probability																			
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Q3	Solve any TWO	20																		
i)	a) Records indicate 1.8 % of the entering students at a university drop out by mid-term. What is the probability that three or fewer students will drop out of a random group of 200 entering students? b) A machine shop repairs small electric motors, which arrive according to a Poisson process at the rate of 12 per week(5 day, 40 hour week). The motors can be repaired on average in 2.5 hours with a variance of 1 hour square. How many working hours should a customer expect to leave a motor at the repair shop?	05 05																		
ii)	a) Six problems are given in an exam. Each problem requires an average of 30 minutes grading time for class of 15 students. The grading time for each problem is exponentially distributed and problems are independent of each other.																			

	i) What is probability that grading will be finished in 2.5 hours or less	03
	ii) What is most likely grading time?	03
	b) The time to failure of a chip follows exponential with a mean of 5000 hours. The chip is in operation for past 1000 hours. What is probability that the chip will be in operation for another 6000 hours.	04
iii)	Write the equations of M/G/1 queuing system. Derive equations for M/M/1 queuing system from M/G/1. Explain steps.	10
Q4	Solve any TWO	20
i)	a) Using K-S test, learn whether the hypothesis that following numbers are uniformly distributed on the interval [0,1] can be rejected. Random numbers: 0.54, 0.05, 0.83, 0.76, 0.48 Use Critical Value: 0.565 for level of significance 0.05	06
	b) What are the limitations of K-S test?	04
ii)	Describe the Acceptance -Rejection Technique for Random Variate Generation. Describe this technique for Poisson random variate generation.	10
iii)	a) Describe properties of random numbers.	04
	b) Why are they called pseudo random numbers?	03
	c) What is the need of random variates?	03
Q5	Solve any FOUR	20
i)	Identifying the distribution of input data	05
ii)	Goodness of fit test for parameter estimation	05
iii)	Calibration of simulation models	05
iv)	Terminating Vs Non terminating simulation	05
v)	Verification of simulation model	05
vi)	Output analysis of steady state simulation	05